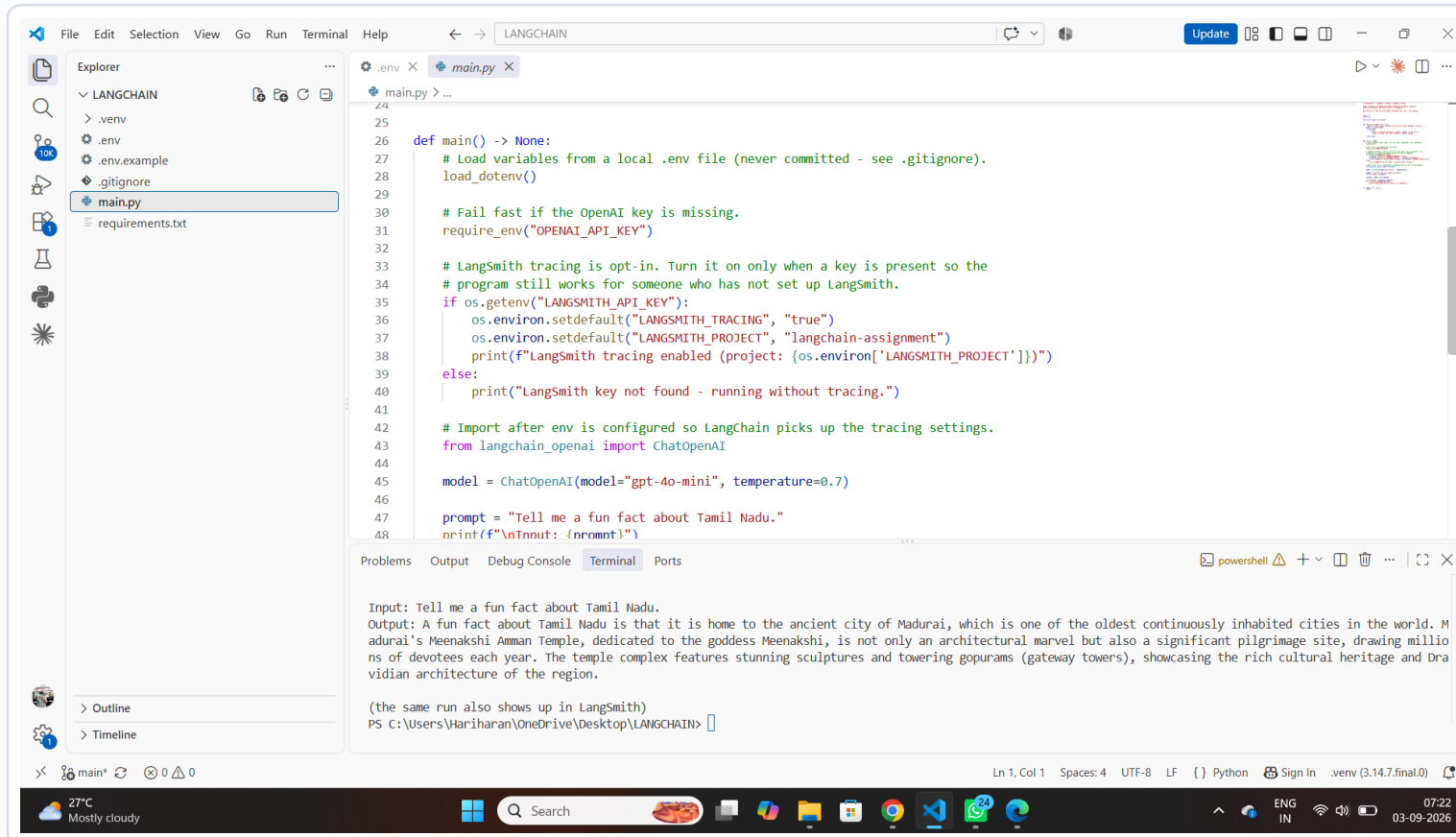


STEP 01

Building the Chain in Python



```
24
25
26 def main() -> None:
27     # Load variables from a local .env file (never committed - see .gitignore).
28     load_dotenv()
29
30     # Fail fast if the OpenAI key is missing.
31     require_env("OPENAI_API_KEY")
32
33     # LangSmith tracing is opt-in. Turn it on only when a key is present so the
34     # program still works for someone who has not set up LangSmith.
35     if os.getenv("LANGSMITH_API_KEY"):
36         os.environ.setdefault("LANGSMITH_TRACING", "true")
37         os.environ.setdefault("LANGSMITH_PROJECT", "langchain-assignment")
38         print(f"LangSmith tracing enabled (project: {os.getenv('LANGSMITH_PROJECT')})")
39     else:
40         print("LangSmith key not found - running without tracing.")
41
42     # Import after env is configured so LangChain picks up the tracing settings.
43     from langchain_openai import ChatOpenAI
44
45     model = ChatOpenAI(model="gpt-4o-mini", temperature=0.7)
46
47     prompt = "Tell me a fun fact about Tamil Nadu."
48     print(f"\nInput: {prompt}")
```

Input: Tell me a fun fact about Tamil Nadu.
Output: A fun fact about Tamil Nadu is that it is home to the ancient city of Madurai, which is one of the oldest continuously inhabited cities in the world. Madurai's Meenakshi Amman Temple, dedicated to the goddess Meenakshi, is not only an architectural marvel but also a significant pilgrimage site, drawing millions of devotees each year. The temple complex features stunning sculptures and towering gopurams (gateway towers), showcasing the rich cultural heritage and Dravidian architecture of the region.

(the same run also shows up in LangSmith)
PS C:\Users\Hariharan\OneDrive\Desktop\LANGCHAIN>

- **Environment first**

load_dotenv() pulls keys from a local .env file, and require_env() fails fast if OPENAI_API_KEY is missing. The .env is git-ignored.

- **Tracing is opt-in**

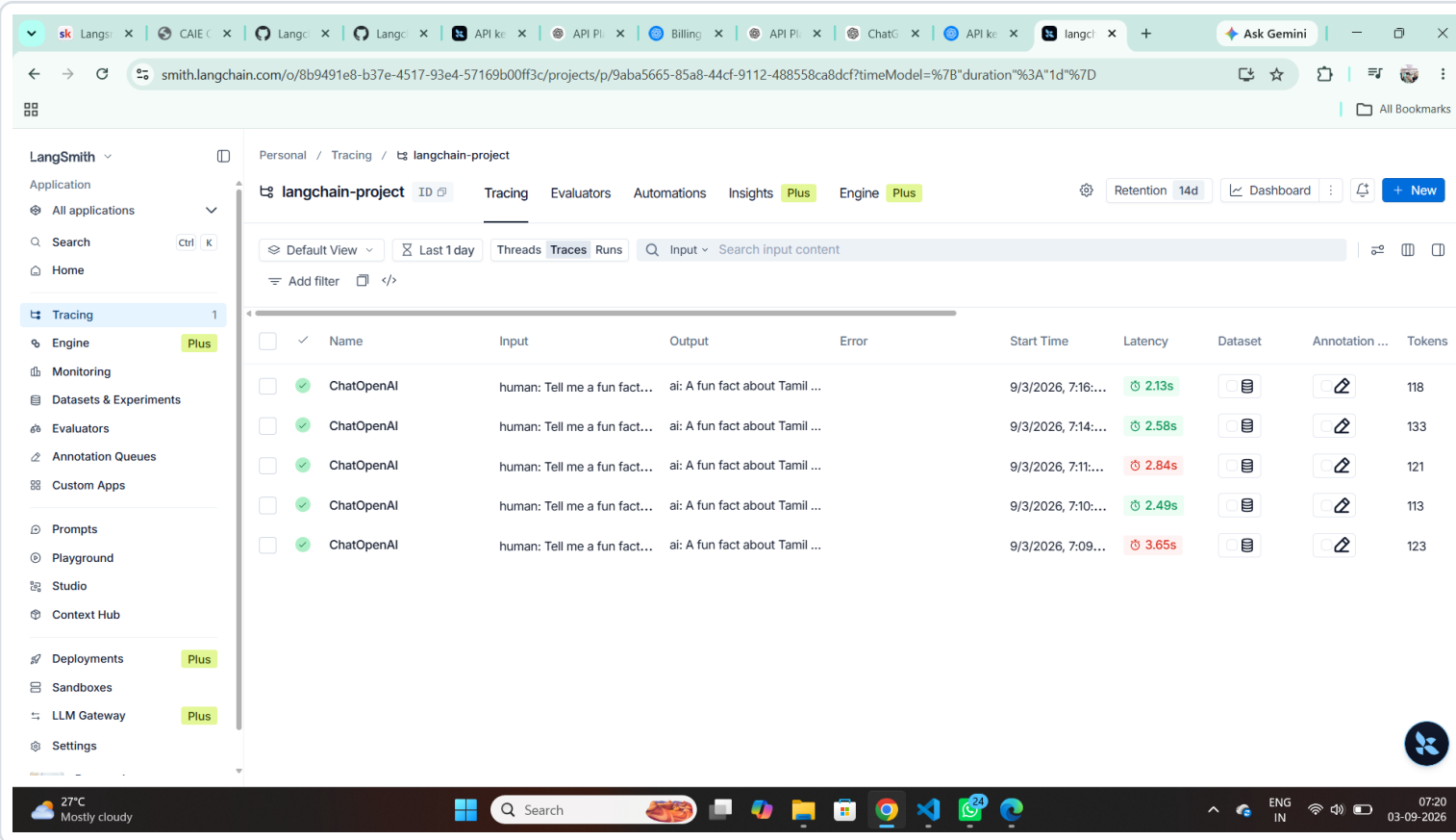
LangSmith switches on only when LANGSMITH_API_KEY exists, so the script still runs for anyone without a LangSmith account.

- **Model call**

ChatOpenAI with gpt-4o-mini at temperature 0.7, prompted for a fun fact about Tamil Nadu. The terminal prints the answer and the run lands in LangSmith.

STEP 02

Runs Captured in LangSmith



The screenshot displays the LangSmith web interface in a browser window. The URL is `smith.langchain.com/o/8b9491e8-b37e-4517-93e4-57169b00ff3c/projects/p/9aba5665-85a8-44cf-9112-488558ca8dcf?timeModel=%7B%22duration%3A%22%7D`. The interface shows the 'Tracing' tab for the 'langchain-project' workspace. A table lists five successful runs, each with a green checkmark icon. The runs are for the 'ChatOpenAI' model, with the input 'human: Tell me a fun fact...' and the output 'ai: A fun fact about Tamil ...'. The latency for each run is displayed in a green box, and the token count is shown in the 'Tokens' column.

	Name	Input	Output	Error	Start Time	Latency	Dataset	Annotation ...	Tokens
☐	☒ ChatOpenAI	human: Tell me a fun fact...	ai: A fun fact about Tamil ...		9/3/2026, 7:16:...	2.13s			118
☐	☒ ChatOpenAI	human: Tell me a fun fact...	ai: A fun fact about Tamil ...		9/3/2026, 7:14:...	2.58s			133
☐	☒ ChatOpenAI	human: Tell me a fun fact...	ai: A fun fact about Tamil ...		9/3/2026, 7:11:...	2.84s			121
☐	☒ ChatOpenAI	human: Tell me a fun fact...	ai: A fun fact about Tamil ...		9/3/2026, 7:10:...	2.49s			113
☐	☒ ChatOpenAI	human: Tell me a fun fact...	ai: A fun fact about Tamil ...		9/3/2026, 7:09:...	3.65s			123

- **Five successful runs**

Every execution between 7:09 and 7:16 AM on 03-09-2026 logged green with no errors, all under the langchain-project workspace.

- **Latency spread**

3.65s on the first cold run, tightening to 2.13s by the last one - the kind of trend that only shows up once runs are traced.

- **Token accounting**

113 to 133 tokens per call, giving a per-run cost baseline before any scaling decisions are made.

STEP 03

Inspecting a Single Trace

The screenshot shows the LangSmith web interface. On the left is a sidebar with a 'Tracing' section containing a list of traces, all labeled 'ChatOpenAI'. The main panel displays a selected trace for 'ChatOpenAI' at 7:16 AM. The trace summary shows 'gpt-4o-mini' with a duration of 2.13s and 118 tokens for less than \$0.0001. The 'Feedback' tab is active, showing a 'User' input: 'Tell me a fun fact about Tamil Nadu.' and an 'AI' output: 'A fun fact about Tamil Nadu is that it is home to the ancient city of Madurai, which is one of the oldest continuously inhabited cities in the world. Madurai's Meenakshi Amman Temple, dedicated to the goddess Meenakshi, is not only an architectural marvel but also a significant pilgrimage site, drawing millions of devotees each year. The temple complex features stunning sculptures and towering gopurams (gateway towers), showcasing the rich cultural heritage and Dravidian architecture of the region.'

- **Full input and output**

The trace stores the exact user prompt alongside the model's reply about Madurai and the Meenakshi Amman Temple - reproducible, not guesswork.

- **Run metadata**

gpt-4o-mini, 2.13s, 118 tokens at under \$0.0001, with Feedback, Input, Output and Attributes tabs on one screen.

- **Why it matters**

Debugging, prompt iteration and cost control all get easier once each call is observable end to end instead of a single print statement.